



Cryptography and Secure Communication

Remote access through a tunnel: WireGuard, OpenVPN and secure configuration of keys, users and allowed networks

Wednesday, October 7, 2026

Cybersecurity **AEC** (Attestation d'études collégiales), program LEA.D8 · Competency EZ03: to apply access control measures

Before we start: the rules of this course

1 Education only

The Material teaches defensive cybersecurity. Nothing in it authorizes access to, tests of, or damage to any real system, network, account or data.

2 Your own isolated laboratory

Practice only in your laboratory (VirtualBox networks and snapshots, or your own free-tier cloud account), or on systems whose owner gave you written permission before you start.

3 The law applies

Unauthorized use of a computer is a criminal offence: Criminal Code of Canada, sections 342.1, 342.2 and 430(1.1). Outside Canada, your local laws apply; college policies apply too.

4 Sign today's acknowledgment

At the start of every session you sign the Acknowledgment and Agreement page. Today: session 16.

You are responsible for your own actions, equipment, accounts and costs. **If an activity seems to touch a system you do not own, stop and ask the instructor.**



Goals and plan of the hour

After this class, you can:

- 1 Explain what a tunnel does to every packet: encapsulation, encryption, integrity protection, endpoint authentication and anti-replay protection.
- 2 Decide whether a company needs a remote-access tunnel, a site-to-site tunnel, or neither.
- 3 Compare WireGuard, OpenVPN and **IPsec** (Internet Protocol Security) on pfSense and OPNsense, and justify a choice in writing.
- 4 Plan secure access: one key pair per device, tunnel addresses, firewall rules, an access register and revocation.

Plan of the hour

0–4	Goals, rules, place in the course
4–22	Lesson 1: what a tunnel does
22–40	Lesson 2: WireGuard, OpenVPN, firewalls
40–56	Lesson 3: keys, users, allowed networks
56–60	LAB_06 preview and questions

Minutes of the class. Videos, code runs, flashcards and practice questions: work period after the break.



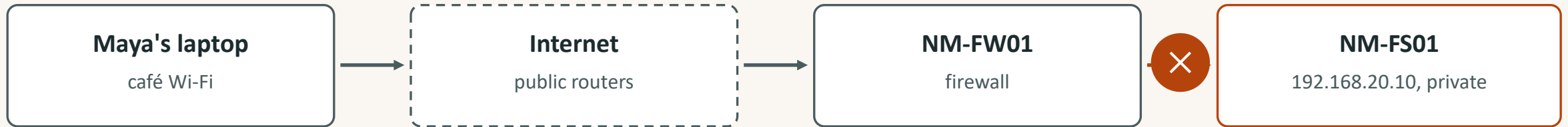
What a tunnel does, and when a company needs one

Three questions for this lesson

- 1 Why can a laptop on café Wi-Fi not reach the file server?
- 2 What does a tunnel do to every packet?
- 3 When is a tunnel the right answer, and when is it not?



A private server and a public path



Internet routers do not forward packets to the private blocks of **RFC** (Request for Comments) 1918: 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16.

- Client records are on NM-FS01 (192.168.20.10, server zone of LAB_01). Not being reachable from the Internet is intended.
- The role table of the Final Project: the remote worker reaches internal data **only** through the tunnel.

1 Design 1: publish the service

A port forward lets every Internet host connect, try passwords and exploit flaws. An unencrypted protocol is readable on the path.

2 Design 2: refuse remote work

Removes the risk and the business benefit at the same time.

3 Design 3: a remote-access tunnel

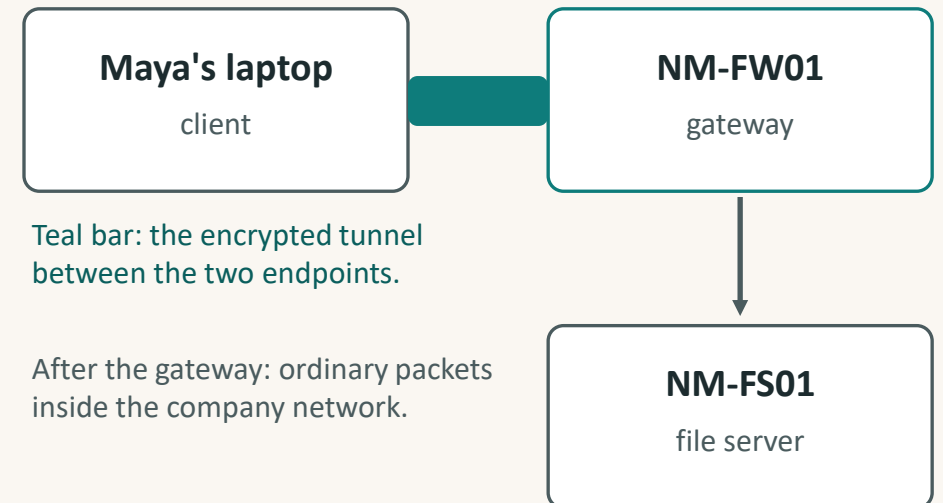
One hardened service admits only authenticated devices; their packets arrive encrypted and then meet the firewall rules.

Virtual private network: the definition

Virtual private network: a logical network built on top of an existing network, usually the Internet, that only authorized endpoints can join. Their traffic crosses the shared network through **tunnels**.

- **Virtual:** no dedicated cable or leased line between the endpoints.
- **Private:** only authorized endpoints can join and read what crosses it.
- **NIST** (National Institute of Standards and Technology) definitions: such a network *can* provide a secure mechanism and is *often* built with encryption and tunneling.
- Encryption is not in every definition: **L2TP** (Layer 2 Tunneling Protocol) creates tunnels without encryption, so it is combined with **IPsec** (Internet Protocol Security).
- In this course every tunnel technology (WireGuard, OpenVPN and the one above) encrypts and authenticates its traffic.

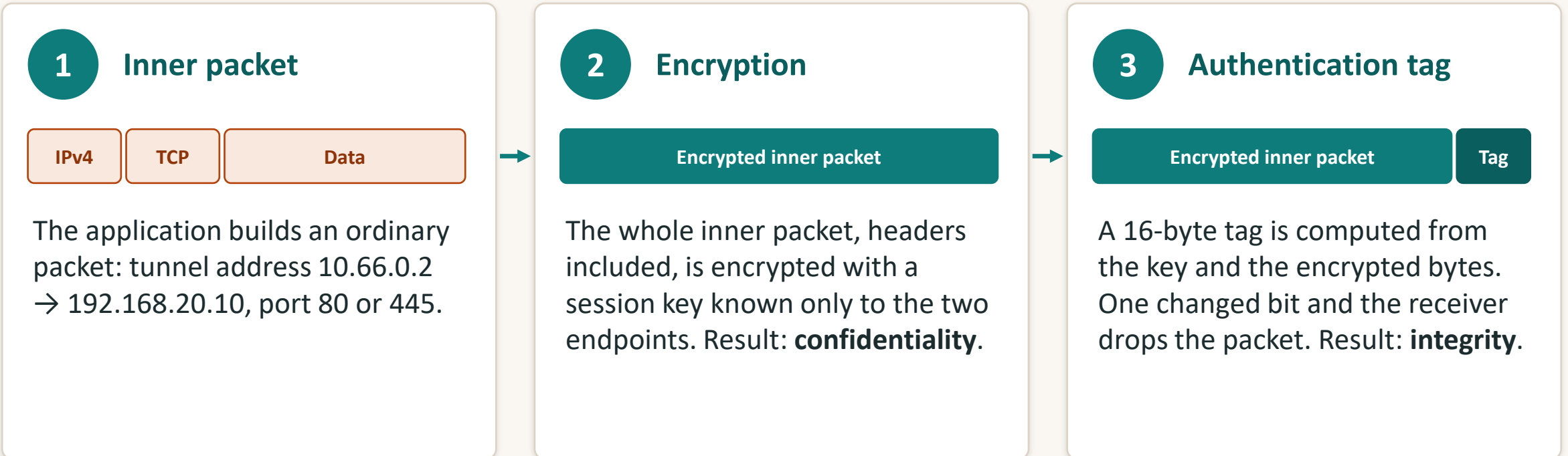
Tunnel endpoints at NorthMaple



Source: https://csrc.nist.gov/glossary/term/virtual_private_network



What the tunnel does to a packet (1/2)

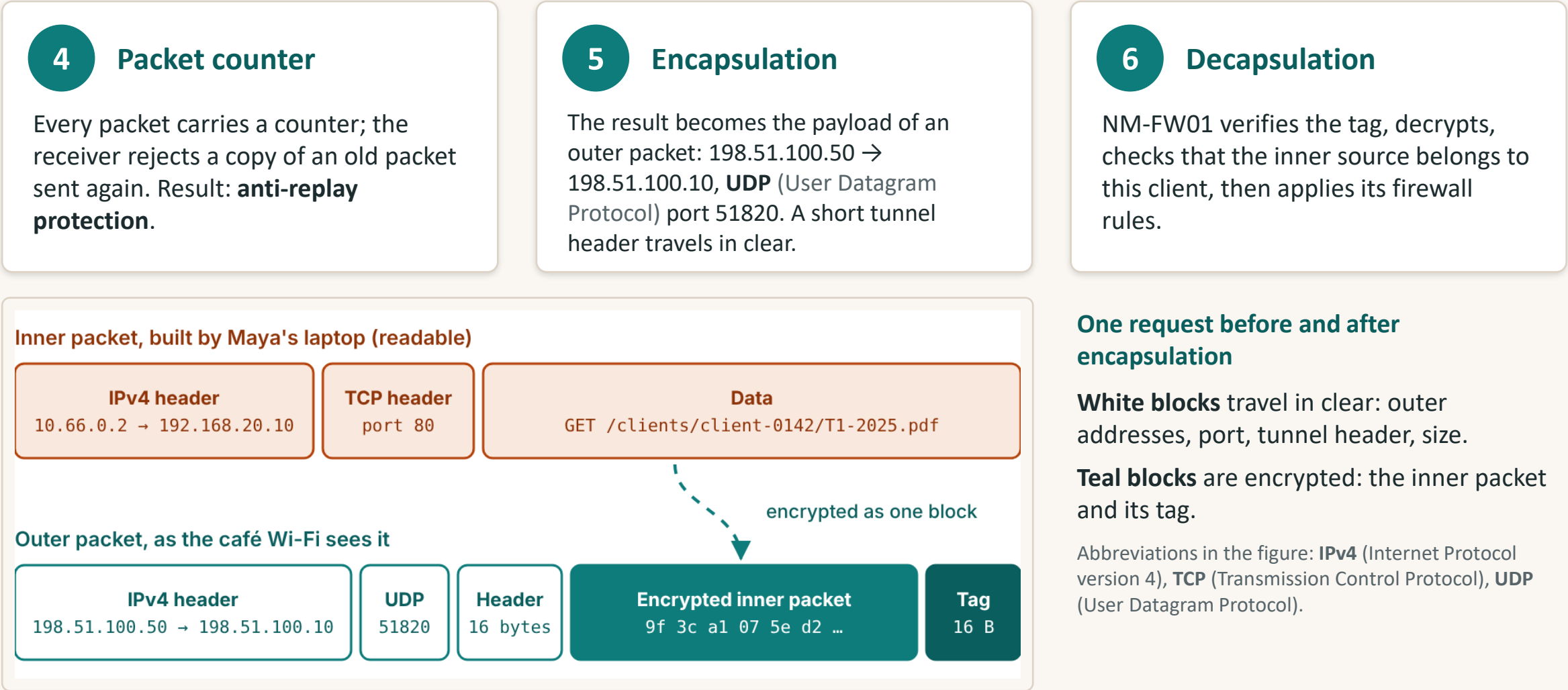


Labels in the strips: **IPv4** (Internet Protocol version 4) header, **TCP** (Transmission Control Protocol) header, data, encrypted bytes, tag.

A cipher that encrypts and computes the tag in one operation is an **AEAD** (Authenticated Encryption with Associated Data) cipher. WireGuard uses ChaCha20-Poly1305; OpenVPN uses **AES-256-GCM** (Advanced Encryption Standard with a 256-bit key in Galois/Counter Mode) by default.

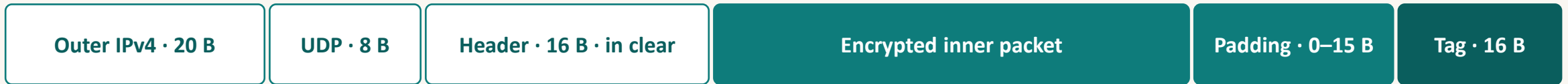


What the tunnel does to a packet (2/2)



Tunnel overhead: what encapsulation costs

One WireGuard packet on the Wi-Fi (not to scale)



In the strip: **IPv4** (Internet Protocol version 4) header, **UDP** (User Datagram Protocol) header, WireGuard header; B = bytes.

60 bytes

fixed overhead over **IPv4** (Internet Protocol version 4)

80 bytes

over **IPv6** (Internet Protocol version 6), whose header is 40 bytes

1420 bytes

usual WireGuard **MTU** (Maximum Transmission Unit): 1500 – 80

- The 16-byte header is **not encrypted**: type (1 byte), reserved (3), receiver index (4), counter (8).
- Before encryption, 0 to 15 zero bytes of **padding** bring the inner packet to a multiple of 16; this hides its exact size and never exceeds the interface **MTU** (Maximum Transmission Unit).
- The code example: 181 bytes inside, padded to 192, **252 bytes** on the Wi-Fi.



Five security properties of a tunnel

Property	What it guarantees	Mechanism	Attack it stops
Confidentiality	Only the two endpoints can read the inner packet	Symmetric encryption with the session key	Passive capture on public Wi-Fi or at an ISP (Internet Service Provider)
Integrity	Any modification is detected	Authentication tag of the cipher	Modification or injection of packets
Endpoint authentication	Each side proves its identity before keys are agreed	Public keys or certificates, optionally user credentials	An unknown device joins; a fake gateway
Anti-replay protection	A captured packet is not accepted twice	Counter in every packet, window of accepted counters	Replay of old packets
Perfect forward secrecy	A stolen long-term key does not decrypt past traffic	Temporary keys; new handshake every 2 minutes (WireGuard)	Decryption of archived captures



What a tunnel does not do

1 The connection stays visible

Outer addresses, port 51820, padded packet sizes and timing stay readable, and the WireGuard header shows which protocol is used.

2 Protection ends at the gateway

After NM-FW01 decrypts a packet, internal controls still apply: zone rules (LAB_05), **NTFS** (New Technology File System) permissions (LAB_02), **HTTPS** (Hypertext Transfer Protocol Secure) on internal services (LAB_06).

3 An infected laptop stays infected

Malware on the remote device can use the tunnel exactly like the user. Host hardening (LAB_03), detection (LAB_04) and least-privilege tunnel rules limit the damage.

4 The gateway becomes a target

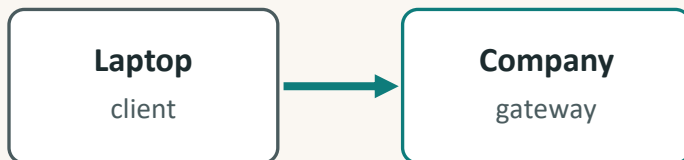
Remote-access gateways are entry points into protected networks and attractive targets (joint **NSA** (National Security Agency) and **CISA** (Cybersecurity and Infrastructure Security Agency) guidance, 2021). Patch the gateway promptly (Class 11).



Three kinds of tunnels

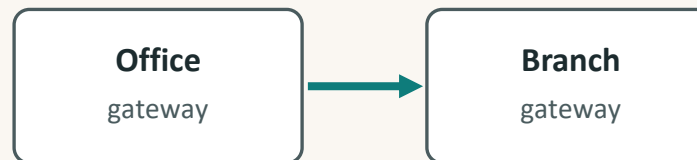
Remote-access

- One device runs a client and holds its own key or certificate.
- It connects that device to the company network.
- **NorthMaple's need.**



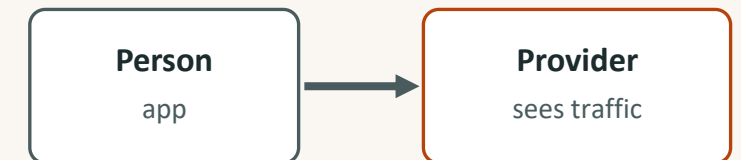
Site-to-site

- Two gateways join two whole networks permanently.
- Head office and branch, or office and a cloud network (Class 13).
- Users run no client.



Commercial privacy service

- Routes one person's Internet traffic through the provider's servers.
- The local network no longer sees it; the provider does.
- No access to any company network.



Full tunnel or split tunnel

Full tunnel

- All traffic of the device goes through the tunnel.
- The client lists 0.0.0.0/0, every **IPv4** (Internet Protocol version 4) address, and ::/0, every **IPv6** (Internet Protocol version 6) address.
- With 0.0.0.0/0 alone, the laptop's **IPv6** (Internet Protocol version 6) traffic leaves outside the tunnel.
- The company can filter everything; office bandwidth is used.

Split tunnel

- Only the listed company networks go through the tunnel, for example 192.168.20.10/32.
- Other traffic goes directly to the Internet.
- The Canadian Centre for Cyber Security advises avoiding split tunnelling where possible: one device then connects the open Internet and the protected tunnel at the same time.

LAB_06 uses a split tunnel because the laboratory is isolated. A production design documents and justifies its choice.

Source: <https://www.cyber.gc.ca/en/guidance/virtual-private-networks-itsap80101>



When does a company need a tunnel?

A company needs a remote-access tunnel, or an equivalent private-access technology, when **one** answer is yes:

1

People outside the office need resources that must not be public

File shares, internal applications, management pages.

2

The information is sensitive or regulated

Québec's private-sector privacy act, amended by Law 25, requires reasonable security measures (authority: the **CAI** (Commission d'accès à l'information du Québec)).

3

Two networks must exchange data continuously

A branch office or a cloud network: then the need is a site-to-site tunnel.

4

The adopted framework requires it

CIS (Center for Internet Security) Safeguard 12.7 (Implementation Groups 2 and 3); Safeguard 6.4, **MFA** (Multi-Factor Authentication) for remote network access (from Group 1).

Course context, not legal advice (Legal Notice, section 7).



When a tunnel is not the answer, and NorthMaple's decision



Public website NM-WEB01 in the DMZ (Demilitarized Zone)

Meant for everyone: protected by **HTTPS** (Hypertext Transfer Protocol Secure), its zone and the firewall rules, not by a tunnel.



Mail service of a cloud provider

TLS (Transport Layer Security) already encrypts it in transit up to the provider, and the provider's sign-in controls access.

NorthMaple's decision: a remote-access tunnel on NM-FW01

Item	Value
Public address of NM-FW01	198.51.100.10
Listen port	UDP (User Datagram Protocol) 51820
Tunnel network	10.66.0.0/24
Firewall, tunnel side	10.66.0.1
Remote worker NM-RW01	10.66.0.2

Lesson 2 chooses the technology. Lesson 3 configures who may use it and what they may reach.



Videos for Lesson 1



VPN (Virtual Private Network) & Remote Working - Computerphile

Computerphile (Dr Steve Bagley) · Length: 14 minutes

<https://www.youtube.com/watch?v=1mtSNVdC7tM>

Explains how a company connects remote staff to internal resources through a **VPN** (Virtual Private Network), which is NorthMaple's exact situation.



Are VPNs (Virtual Private Networks) Just A Scam?

Techquickie · Length: 6 minutes

<https://www.youtube.com/watch?v=239w7x2TdWE>

Shows what commercial privacy **VPN** (Virtual Private Network) services can and cannot do, so students do not confuse them with a company remote-access **VPN** (Virtual Private Network).

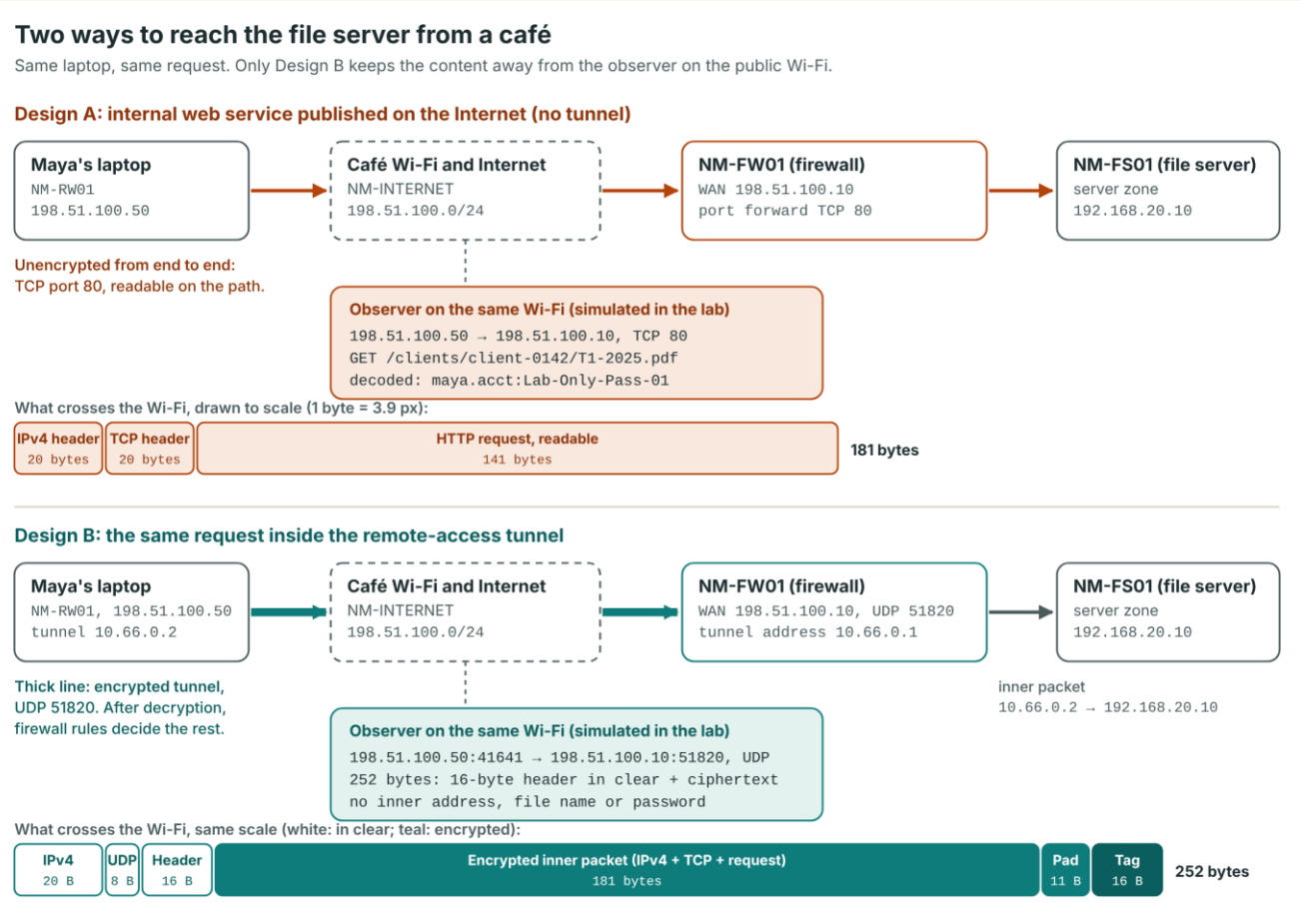
Task for the work period: watch both videos (each under 15 minutes), then explain in two sentences the difference between a company remote-access tunnel and a commercial privacy service.

TikTok: no direct links. Search #vpnexplained, #remotework, #cybersecurity. Keep a clip only if the creator is identifiable, the versions are recent, the demonstration uses the creator's own equipment, and every claim matches the official documentation.

Check any link first with the VirusTotal scanner: <https://www.virustotal.com/gui/home/url> (Legal Notice, section 10). Public clips are laboratory demonstrations: do not reproduce them outside a laboratory (section 12).



Maya at the café: Design A and Design B



Design A: port forward, TCP (Transmission Control Protocol) **80**

The observer reads the file path, the client number and, after decoding Base64, the password.

Design B: WireGuard tunnel

Two addresses, port 51820 and a 252-byte packet: header readable, content not. A modified packet is rejected.

On a real hotspot the observer would see the laptop's private address behind the café's **NAT** (Network Address Translation) router.

Never capture traffic on a network you do not own (Legal Notice, section 3).

Figure 1.1: same request, same byte scale. Abbreviations in the figure: **IPv4** (Internet Protocol version 4), **TCP** (Transmission Control Protocol), **UDP** (User Datagram Protocol), **HTTP** (Hypertext Transfer Protocol).



Code demo: the Tunnel Observer

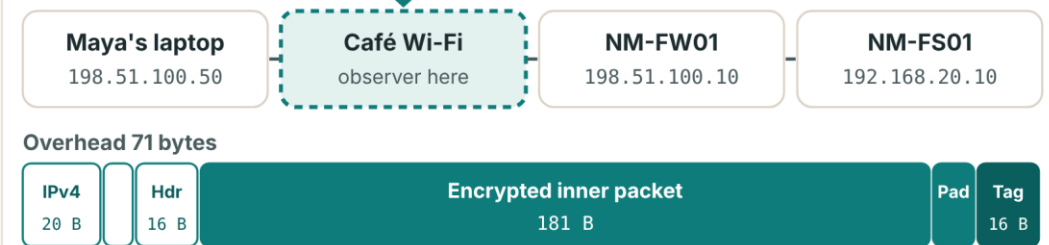
- A self-contained page builds Maya's request as real bytes, then encrypts it as a tunnel would.
- It pads the inner packet to a multiple of 16 bytes, encrypts it with **AES-256-GCM** (Advanced Encryption Standard with a 256-bit key in Galois/Counter Mode) and adds a WireGuard-style header.
- One changed bit: the gateway rejects the packet.

```
const padded = new Uint8Array(
  Math.ceil(inner.length / 16) * 16);
padded.set(inner, 0);
const ciphertext = await crypto.subtle.encrypt(
  { name: "AES-GCM", iv: nonce }, key, padded);
```

Output (Design B)

```
Observer sees: ... -> 198.51.100.10:51820, UDP, 252 bytes
Tunnel overhead: 71 bytes (... + 11 padding)
Modified packet rejected: tag check failed.
```

Design B: remote-access tunnel



What the observer on the café Wi-Fi reads

```
198.51.100.50:41641 → 198.51.100.10:51820, UDP
252 bytes
header (clear): 04 00 00 00 2a 00 00 00 00 00 00 00 00 00 00 00
= type 4 (data), receiver index 42, counter 0
ciphertext: 18 93 ca d9 9c 31 d4 de 59 e5 49 7d 2e 54 20 e4 f3 07 ...
no inner address, file name or password
```

Runner view at step 28 of 34. File: 420-D03-AB_C16_L1_Tunnel_Observer.html.
Abbreviations in the view: **IPv4** (Internet Protocol version 4), **TCP** (Transmission Control Protocol), **HTTP** (Hypertext Transfer Protocol), **UDP** (User Datagram Protocol).



Encapsulation, step by step

Step 1 of 7

Maya's laptop builds the inner packet. Every field is still readable.



the packet is here



- 1 Build the inner packet
- 2 Pad and encrypt it
- 3 Compute the tag
- 4 Add the tunnel and outer headers
- 5 Cross the café Wi-Fi
- 6 Check, decrypt, remove padding
- 7 Forward to NM-FS01

Figure 1.2: seven steps of about 2.6 seconds; the picture moves in slide-show mode. Abbreviations in the animation: **IPv4** (Internet Protocol version 4), **TCP** (Transmission Control Protocol), **UDP** (User Datagram Protocol).



WireGuard, OpenVPN and the firewall's built-in options

Three questions for this lesson

- 1 How do WireGuard and OpenVPN differ?
- 2 What do pfSense and OPNsense provide without an extra server?
- 3 How do we choose, and write the choice down?



Free and open-source: what it gives, what it does not remove

What it gives

- Source code published under a licence that allows reading, testing, modifying and redistributing.
- Independent review reduces the chance that a weakness stays hidden.
- No licence fee per user: the cost is the administrator's time.

What it does not remove

- Installing updates.
- Configuring the product correctly.
- Monitoring it.
- These remain the company's work.

pfSense CE (Community Edition) 2.9.0 · Netgate

Free and open-source firewall and router.

OPNsense 26.7 · Deciso

Simplified two-clause **BSD** (Berkeley Software Distribution) licence.

Product names are trademarks of their owners, used for identification only; Cyber.SoHo has no affiliation with Netgate, Deciso, WireGuard or OpenVPN (Legal Notice, section 9).



WireGuard (1/2): four design decisions

Designed by Jason A. Donenfeld; peer-reviewed paper in 2017; part of the Linux kernel since version 5.6 (2020).

1 A small implementation

Under 4,000 lines of code for Linux, without the cryptographic primitives: one person can read and audit all of it.

2 One fixed set of algorithms

No algorithm negotiation: the peers never discuss ciphers, so an attacker cannot push them toward a weaker one.

3 Identity is a public key

Each peer is identified only by its public key and linked to its allowed addresses: cryptokey routing (Lesson 3).

4 No answer to unauthenticated packets

A scan of 198.51.100.10 on **UDP** (User Datagram Protocol) 51820 receives no reply, and no state is created for unknown senders.



WireGuard (2/2): algorithms, timers and limits

Algorithms (fixed)

- Curve25519: key agreement with **ECDH** (Elliptic-Curve Diffie-Hellman)
- ChaCha20-Poly1305: encryption and integrity
- BLAKE2s: hashing
- **HKDF** (Key Derivation Function built on a Hash-based Message Authentication Code): key derivation
- SipHash24: internal tables
- Handshake: Noise IK pattern

Timers

- New handshake every **120 seconds** while traffic flows.
- Keys refused after **180 seconds**.
- Result: **PFS** (Perfect Forward Secrecy).

Limits (Netgate)

- No connections or sessions.
- No user authentication: no user name, password or second factor.
- Minimal logging.
- Static tunnel addresses, set by hand for every peer.

UDP (User Datagram Protocol) only. Port 51820 is a convention and the default of pfSense and OPNsense, not an **IANA** (Internet Assigned Numbers Authority) assignment.



OpenVPN (1/2): two channels

First released by James Yonan in 2001. Port 1194 over **UDP** (User Datagram Protocol) or **TCP** (Transmission Control Protocol), registered with **IANA** (Internet Assigned Numbers Authority).

C

Control channel: a TLS (Transport Layer Security) session

The peers authenticate each other with X.509 certificates from the company **CA** (Certificate Authority) (Class 15) and agree on session keys.

D

Data channel: the encrypted packets

Default data ciphers in version 2.6: **AES-256-GCM** (Advanced Encryption Standard with a 256-bit key in Galois/Counter Mode), **AES-128-GCM** (Advanced Encryption Standard with a 128-bit key in Galois/Counter Mode) and CHACHA20-POLY1305, all **AEAD** (Authenticated Encryption with Associated Data) ciphers.

Source: <https://build.openvpn.net/man/openvpn-2.6/openvpn.8.html>



OpenVPN (2/2): options, authentication and costs

Three named options

- `tls-crypt`: every control-channel packet is encrypted and authenticated with a pre-shared key; the server ignores everyone else.
- `dev tun` carries **IP** (Internet Protocol) packets (layer 3); `dev tap` carries Ethernet frames (layer 2).
- **DCO** (Data Channel Offload) moves data encryption into the kernel.

Strength: authentication

- Certificate, or user name and password checked locally, by **LDAP** (Lightweight Directory Access Protocol) or by **RADIUS** (Remote Authentication Dial-In User Service), or both.
- Netgate calls certificate plus password the most secure combination.
- OPNsense adds **TOTP** (Time-based One-Time Password) codes: real **MFA** (Multi-Factor Authentication).

Costs

- **TCP** (Transmission Control Protocol) port 443 works where only web traffic is allowed, but a tunnel inside **TCP** (Transmission Control Protocol) retransmits lost data twice: slower.
- Certificates must be issued, renewed and revoked.
- Weak legacy settings remain selectable.



Built into the firewalls (1/2): pfSense and OPNsense

Netgate lists four tunnel types for pfSense: **IPsec** (Internet Protocol Security), OpenVPN, WireGuard and **L2TP** (Layer 2 Tunneling Protocol).

pfSense CE (Community Edition) 2.9.0

- **IPsec** (Internet Protocol Security) and OpenVPN built in; OpenVPN wizard; certificates in System > Certificates; optional client export package.
- WireGuard: add-on package from System > Package Manager, then configured under VPN > WireGuard (tunnels and peers).
- Default listen port 51820; the page generates the keys.

OPNsense 26.7

- **IPsec** (Internet Protocol Security), OpenVPN and WireGuard are all part of the core system.
- WireGuard moved from a plugin into the core with version 24.1.
- Guide "WireGuard Road Warrior Setup": instance (port 51820, tunnel address), one peer per client with a /32 address, a **WAN** (Wide Area Network) rule, rules on the WireGuard interface.



Built into the firewalls (2/2): the standards option, and what to avoid

IPsec (Internet Protocol Security) **with IKEv2** (Internet Key Exchange version 2)

- **IKE** (Internet Key Exchange) on **UDP** (User Datagram Protocol) 500; **NAT** (Network Address Translation) traversal on **UDP** (User Datagram Protocol) 4500; data in **ESP** (Encapsulating Security Payload).
- Defined in **RFC** (Request for Comments) 4301 and **RFC** (Request for Comments) 7296; studied in Class 11.
- Clients built into Windows, macOS, iOS and Android.
- Many parameters must match on both sides.

 **L2TP** (Layer 2 Tunneling Protocol) **alone**

Provides no encryption by itself; it is combined with **IPsec** (Internet Protocol Security).

 **PPTP** (Point-to-Point Tunneling Protocol): **do not use**

Netgate: its authentication is broken and captured traffic can always be decrypted. pfSense includes no server for it.

Side-by-side comparison (1/2)

Attribute	WireGuard	OpenVPN	IPsec (Internet Protocol Security) with IKEv2 (Internet Key Exchange version 2)
Transport and port	UDP (User Datagram Protocol) only; 51820 by convention	UDP (User Datagram Protocol) or TCP (Transmission Control Protocol); 1194, registered	UDP (User Datagram Protocol) 500 and 4500
Algorithms	Fixed: Curve25519, ChaCha20-Poly1305, BLAKE2s	Control channel in TLS (Transport Layer Security); AES-256-GCM (Advanced Encryption Standard with a 256-bit key in Galois/Counter Mode) data by default	Negotiated proposals
Algorithm negotiation	None	Yes	Yes
Identity of a device	Public key	Certificate from the company CA (Certificate Authority)	Certificate or EAP (Extensible Authentication Protocol) credentials
User authentication and MFA (Multi-Factor Authentication)	None	Password checked locally, by LDAP (Lightweight Directory Access Protocol) or RADIUS (Remote Authentication Dial-In User Service); TOTP (Time-based One-Time Password) on OPNsense	Credentials on the firewall or a RADIUS (Remote Authentication Dial-In User Service) server, which can add a second factor



Side-by-side comparison (2/2)

Attribute	WireGuard	OpenVPN	The standards option
Answer to unauthenticated packets	None	None when <code>tls-crypt</code> is used	Answers IKE (Internet Key Exchange) requests
Code size	Under 4,000 lines (Linux)	Large (OpenVPN and OpenSSL)	Large
pfSense CE (Community Edition) 2.9.0	Add-on package	Built in, with wizard	Built in
OPNsense 26.7	Core system	Built in	Built in
Clients	Official apps, to install	Clients to install	Built into major systems

Netgate rates all three as **cryptographically secure when properly configured**. The standards option is **IPsec** (Internet Protocol Security) with **IKEv2** (Internet Key Exchange version 2).

Sources: <https://docs.netgate.com/pfsense/en/latest/vpn/selection.html>, the WireGuard protocol page and the OpenVPN 2.6 manual.



What official guidance says

Canadian Centre for Cyber Security

- **IPsec** (Internet Protocol Security) as the first option
- Phishing-resistant **MFA** (Multi-Factor Authentication)
- Current patches
- Avoid split tunnelling

NSA (National Security Agency) and CISA (Cybersecurity and Infrastructure Security Agency), 2021

- Standards-based products validated by **NIAP** (National Information Assurance Partnership)
- Require **MFA** (Multi-Factor Authentication)
- Patch quickly
- Disable unneeded features

NIST (National Institute of Standards and Technology)

- Special Publication 800-77 Revision 1: guide to the standards option
- Special Publication 800-46 Revision 2: telework security

These documents do not contradict WireGuard for LAB_06: **the best technology depends on the organization's requirements.**



Choosing with a documented method

1 Mandatory requirements

Pass/fail. One failure excludes the option before any score.

2 Weights

Each desirable criterion gets a weight; the weights add up to 100 %.

3 Scores

1 to 5 per criterion, each with one sentence of evidence.

4 Totals

Total = sum of (weight × score).

5 Sensitivity analysis

Change plausible weights or requirements: does the winner change?

6 Decision record

Date, versions, requirements, scores, result, accepted limitation.

$$\text{Total} = \Sigma (\text{weight} \times \text{score})$$



Videos for Lesson 2



Is WIREGUARD Over-Hyped (Vs OpenVPN)?

apalrd's adventures · Length: under 3 minutes

<https://www.youtube.com/shorts/8cLsUI2VJyo>

A short, critical comparison of the two protocols of this lesson.



What is Wireguard? A "New" VPN (Virtual Private Network) Protocol + How it Compares to OpenVPN

All Things Secured · Length: check on YouTube

<https://www.youtube.com/watch?v=mxpHRdO4rDU>

A beginner-level introduction to WireGuard and a side-by-side comparison with OpenVPN.



pfSense OpenVPN VS WireGuard

Lawrence Systems · Length: check on YouTube

<https://www.youtube.com/watch?v=gH3FI1h-AYE>

The two options as they appear in pfSense, the firewall used in LAB_05 and LAB_06.



TCP (Transmission Control Protocol) Meltdown - Computerphile

Computerphile · Length: about 15 minutes

https://www.youtube.com/watch?v=AAask2N_oPk

Optional: why tunnels carried over **TCP** (Transmission Control Protocol) perform badly, which explains WireGuard's **UDP** (User Datagram Protocol)-only design and OpenVPN's **UDP** (User Datagram Protocol) default.

TikTok: no direct links. Search #wireguard, #openvpn, #pfsense, #opnsense. Keep a clip only if the creator is identifiable, the versions are recent, the demonstration uses the creator's own equipment, and every claim matches the official documentation.

Check any link first with the VirusTotal scanner: <https://www.virustotal.com/gui/home/url> (Legal Notice, section 10). Public clips are laboratory demonstrations: do not reproduce them outside a laboratory (section 12).



NorthMaple's requirements and scores

Mandatory requirements

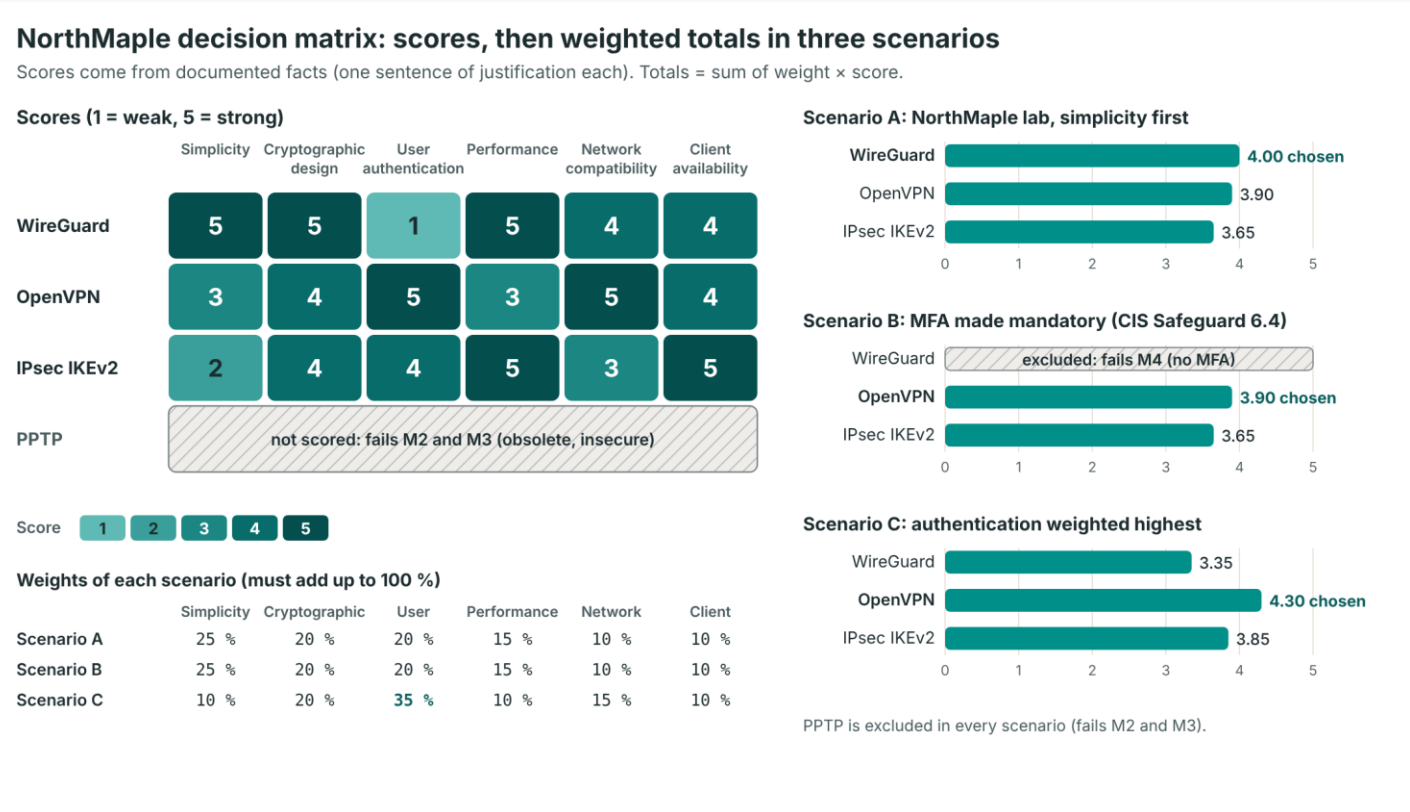
- M1** Free and open-source
- M2** Available on pfSense **CE** (Community Edition) 2.9.0 and OPNsense 26.7
- M3** Modern cryptography, not known to be broken
- M4 MFA** (Multi-Factor Authentication) through the firewall's authentication services (scenario B only)
- PPTP** (Point-to-Point Tunneling Protocol) **fails M2 and M3: excluded, never scored.**

Criterion (weight in A)	WireGuard	OpenVPN	Standards option
Configuration simplicity (25 %)	5	3	2
Cryptographic design (20 %)	5	4	4
User authentication options (20 %)	1	5	4
Performance (15 %)	5	3	5
Network compatibility (10 %)	4	5	3
Client availability (10 %)	4	4	5
Weighted total, scenario A	4.00	3.90	3.65

Scores from 1 (weak) to 5 (strong). Standards option: **IPsec** (Internet Protocol Security) with **IKEv2** (Internet Key Exchange version 2).



Three scenarios, three answers



A · simplicity first (laboratory)

WireGuard 4.00, ahead of OpenVPN by 0.10

B · MFA (Multi-Factor Authentication) mandatory

WireGuard excluded by M4; OpenVPN 3.90

C · authentication weighted 35 %

OpenVPN 4.30; WireGuard last with 3.35

Figure 2.1. Abbreviations in the figure: **MFA** (Multi-Factor Authentication), **CIS** (Center for Internet Security), **IPsec** (Internet Protocol Security), **IKEv2** (Internet Key Exchange version 2), **PPTP** (Point-to-Point Tunneling Protocol).

Course decision: WireGuard for LAB_06. Accepted limitation: no native **MFA** (Multi-Factor Authentication), compensated by one key per device, strict tunnel rules, a 90-day register review and monitoring by the **IDS** (Intrusion Detection System) of LAB_04. Real client data in production would need it.

Code demo: the decision matrix in Python

- Excludes every option that fails a mandatory requirement, then computes and ranks the weighted totals.
- Stops if the weights of a scenario do not add up to 1.00.
- Writes every result to a **CSV** (Comma-Separated Values) file for the decision record.

```
def weighted_total(scores, weights):  
    return round(sum(s * w for s, w  
                    in zip(scores, weights)), 2)  
  
assert abs(sum(weights) - 1.0) < 1e-9
```

Output (end of scenario C)

```
1. OpenVPN          4.30 #####...  
DECISION: OpenVPN (margin 0.45 over IPsec IKEv2)
```

Scores (1 = weak, 5 = strong)

	Weight	WireGuard	OpenVPN	IPsec IKEv2
Simplicity		5	3	2
Cryptographic design		5	4	4
User authentication		1	5	4
Performance		5	3	5
Network compatibility		4	5	3
Client availability		4	4	5

Weighted total

PPTP: not scored, fails M2 and M3 in every scenario

Scenario A: simplicity first

WireGuard 4.00 chosen

OpenVPN 3.90

IPsec IKEv2 3.65

Scenario B: MFA mandatory

WireGuard excluded

OpenVPN 3.90 chosen

IPsec IKEv2 3.65

Scenario C: authentication 35 %

WireGuard 3.35

OpenVPN 4.30 chosen

IPsec IKEv2 3.85

Computation

All three scenarios computed. Results are written to vpn_decision_matrix.csv.

Runner view after the last step. File: 420-D03-AB_C16_L2_VPN_Decision_Matrix.py.
Abbreviations in the view and output: **MFA** (Multi-Factor Authentication), **IPsec** (Internet Protocol Security), **IKEv2** (Internet Key Exchange version 2), **PPTP** (Point-to-Point Tunneling Protocol).

Keys, users and allowed networks

Three questions for this lesson

- 1 Who is connecting, and how do we know?
- 2 What may each device reach, and who enforces it?
- 3 What happens when a laptop is stolen?



Three questions for every remote connection

The same three questions as on a file server (Classes 3 and 4), together called **AAA** (Authentication, Authorization and Accounting):

1 Authentication

Who is connecting?

A key, a certificate, a password, a second factor, or a combination of them.

2 Authorization

What may this identity reach?

The allowed networks and the firewall rules of the tunnel interface.

3 Accounting

What happened, and when?

The logs (handshake times, firewall decisions) and the access register.

CIS (Center for Internet Security) Safeguard 12.7 (Implementation Groups 2 and 3): users authenticate to enterprise-managed tunnel and authentication services before reaching enterprise resources. Safeguard 6.4 adds **MFA** (Multi-Factor Authentication) (from Group 1).



Keys: one key pair per device

- A WireGuard identity is a **key pair** on Curve25519, the curve of **RFC** (Request for Comments) 7748.
- **Private key**: 32 random bytes that never leave the device.
- **Public key**: computed from it with X25519; safe to show to anyone. Both are 44 Base64 characters.

Official commands

```
umask 077                # owner-only files
wg genkey > privatekey    # 32 random bytes
wg pubkey < privatekey > publickey # derived public key
```

During the handshake

- Each side combines its own private key with the other side's public key: both get the same shared secret without sending it. This is key agreement with **ECDH** (Elliptic-Curve Diffie-Hellman).
- Fresh temporary keys for every handshake: session keys change every two minutes, so **PFS** (Perfect Forward Secrecy) holds.
- Optional **PSK** (Pre-Shared Key) per peer: Netgate presents it as a layer of post-quantum resistance.



Six rules of key hygiene

1 One key pair per device

Never share a key between two people or two devices.

2 Generate it on the device

Send only the public key to the administrator.

3 Never send a private key

Not by e-mail or chat, and never into an **AI** (Artificial Intelligence) tool.

4 Protect key files

Permissions 600 (owner only), given by `umask 077`.

5 Record every public key

With its owner, device and tunnel address, in the access register.

6 Revoke at once

Lost, stolen, replaced or retired: revoke; the new device gets a new key pair.

OpenVPN uses certificates: one per user or device, signed by the company **CA** (Certificate Authority); revocation goes through the **CRL** (Certificate Revocation List).



Users: WireGuard knows peers, not people

- NM-FW01 sees a public key, never a name: the link between a key and a person exists only in the administrator's records.
- That is why the **access register** is mandatory: one line per device.
- Accounts and keys are personal: a shared account breaks accountability.

One line of the access register

Field	Example
Person and role	maya.acct, Accountant
Device	laptop-acc-02
Public key	jZoZbe++osG...
Tunnel address	10.66.0.2
Issued, review	issue date; review after 90 days
Status	active, suspended or revoked

Technologies that authenticate people

- **OpenVPN**: user name and password checked locally, by **LDAP** (Lightweight Directory Access Protocol) or by **RADIUS** (Remote Authentication Dial-In User Service); on OPNsense, a **TOTP** (Time-based One-Time Password) code can be added.
- **IPsec** (Internet Protocol Security): **EAP** (Extensible Authentication Protocol) credentials stored on the firewall or checked by a **RADIUS** (Remote Authentication Dial-In User Service) server, which can add a second factor.
- A certificate or a code plus a password is **MFA** (Multi-Factor Authentication): something you have and something you know.



Allowed networks (1/2): AllowedIPs on NM-FW01

- Tunnel network 10.66.0.0/24: NM-FW01 is 10.66.0.1, then 10.66.0.2, 10.66.0.3... Each device gets a **unique** tunnel address.
- For each peer, NM-FW01 stores AllowedIPs with exactly that peer's address and a /32 prefix. The list works in **both directions**:



Incoming packets: an access list

After decrypting with peer X's key, NM-FW01 accepts the packet only if its inner source belongs to X's list. A laptop that claims another device's address is rejected.



Outgoing packets: a routing table

A reply to 10.66.0.3 is encrypted for the peer whose list contains 10.66.0.3, and for no one else.

Peer (public key)	AllowedIPs on NM-FW01
laptop-acc-02 · jZoZbe++osG...	10.66.0.2/32
laptop-tech-01 · E1hvdIbG00M...	10.66.0.3/32



Allowed networks (2/2): the client side and least privilege

On the client

- Same two rules toward its only peer, NM-FW01: routes outgoing destinations, filters incoming sources.
- 192.168.20.10/32: narrow split tunnel.
- 0.0.0.0/0 with ::/0: full tunnel.
- The user can edit it, so it **cannot enforce** the company's policy.

Least privilege, enforced on the WireGuard interface of NM-FW01

#	Source	Destination and port	Action
1	10.66.0.2 (accountant)	192.168.20.10, SMB (Server Message Block) on TCP (Transmission Control Protocol) 445	pass
2	10.66.0.3 (technician)	192.168.20.10, SMB (Server Message Block) on TCP (Transmission Control Protocol) 445	pass
3	10.66.0.3	192.168.10.0/24, SSH (Secure Shell) on 22, RDP (Remote Desktop Protocol) on 3389	pass
4	10.66.0.0/24	any	block, logged

Rule 4 is the **default deny**. Both roles reach port 445, but the **NTFS** (New Technology File System) permissions of LAB_02 decide which folders each one opens: two layers.



Issue, review, revoke

1 Issue

- The device generates its key pair and sends the public key.
- Next free tunnel address; the role's firewall rules.
- Register line written; connection tested.



2 Review

- Every 90 days and at every change of role.
- Compare peers on NM-FW01 with active register lines.
- Check last handshake times; remove unused peers.



3 Revoke

- Lost, stolen, retired device or departure: delete or disable the peer at once.
- Mark the register line revoked, with the date.
- Check the logs for use after the loss.

Final Project resilience rows: **2** password guessing (no password to guess, no answer to unauthenticated packets) · **5** lost laptop (revoke the key; disk encryption of LAB_06 part 2) · **8** accountant tries technicians' data (tunnel rules and **NTFS** (New Technology File System) both refuse).



Videos for Lesson 3



Public Key Cryptography - Computerphile

Computerphile (Robert Miles) · Length: 6 minutes

https://www.youtube.com/watch?v=GSIDS_lvRv4

The roles of the private key and the public key, the basis of every WireGuard peer.



Secret Key Exchange (Diffie-Hellman) - Computerphile

Computerphile · Length: 9 minutes

<https://www.youtube.com/watch?v=NmM9HA2MQGI>

How two peers compute the same session key without ever sending it, the idea behind the WireGuard handshake.



Elliptic Curves - Computerphile

Computerphile · Length: 9 minutes

<https://www.youtube.com/watch?v=NF1pwjL9-DE>

Background on elliptic-curve cryptography, the family of Curve25519 used for WireGuard keys.



How WireGuard's Cryptokey Routing Works

Hacker Workspace · Length: under 3 minutes

<https://www.youtube.com/shorts/OTkwTYSZWfO>

AllowedIPs as both a routing table and an access list, in under three minutes.

TikTok: no direct links. Search #wireguard, #cryptography, #networksecurity. Keep a clip only if the creator is identifiable, the versions are recent, the demonstration uses the creator's own equipment, and every claim matches the official documentation.

Check any link first with the VirusTotal scanner: <https://www.virustotal.com/gui/home/url> (Legal Notice, section 10). Public clips are laboratory demonstrations: do not reproduce them outside a laboratory (section 12).

A new remote worker and a stolen laptop

Device	User and role	Status	Tunnel address	Allowed destinations
laptop-acc-02	maya.acct, Accountant	active (new key pair)	10.66.0.2	192.168.20.10 TCP (Transmission Control Protocol) 445
laptop-tech-01	leo.tech, Technician	active	10.66.0.3	192.168.20.10 port 445; 192.168.10.0/24 ports 22 and 3389
laptop-acc-01	maya.acct, Accountant	revoked (stolen)	none	none: peer removed from NM-FW01

Access test: five connections

ALLOW	Maya's new laptop → client-records share	DENY	Maya's new laptop → remote desktop on NM-WS01
ALLOW	Leo's laptop → remote desktop on NM-WS01	DENY	Leo's laptop → management page of NM-FW01
REJECT	Stolen laptop, old key → file server: unknown key, no answer		



Code demo: building the access register

- Generates one key pair per active device, assigns tunnel addresses and writes the register and the [Peer] blocks.
- A revoked device gets no key and no peer.
- Prints ALLOW, DENY or REJECT for five test connections.

```
set -euo pipefail
umask 077
wg genkey > "$KEYS/$DEVICE.key"
PUB=$(wg pubkey < "$KEYS/$DEVICE.key")
printf '[Peer]\nPublicKey = %s\n' "$PUB" ...
```

Output (access test)

```
ALLOW laptop-acc-02 -> 192.168.20.10:445
REJECT laptop-acc-01 -> 192.168.20.10:445
```

Result: 2 ALLOW, 2 DENY, 1 REJECT

Remote devices

laptop-acc-02
maya.acct, Accountant
tunnel 10.66.0.2
pub HRJv/gKvdXqdpFTi07ewLHADewGvoD1...

laptop-tech-01
leo.tech, Technician
tunnel 10.66.0.3
pub LakUGXjwvyHCe2nJf4I9JrYx...

laptop-acc-01
maya.acct, Accountant
revoked: no key, no peer

NM-FW01 peers (cryptokey routing)

PublicKey	AllowedIPs
HRJv/gKvdXqdpFTi07ewLHADewGvoD1...	10.66.0.2/32
LakUGXjwvyHCe2nJf4I9JrYxpm2v7ot...	10.66.0.3/32

A key that is not listed here gets no answer.

Rules of the tunnel interface
Accountant: 192.168.20.10 TCP 445
Technician: 192.168.20.10 TCP 445
Technician: 192.168.10.0/24 TCP 22, 3389
Everything else: default deny

Access test

	NM-FS01:445	NM-WS01:3389	NM-FW01:443
laptop-acc-02	✓ ALLOW	✗ DENY	not tested
laptop-tech-01	not tested	✓ ALLOW	✗ DENY
laptop-acc-01	⌚ REJECT	not tested	not tested

Runner view after the last step. File: 420-D03-AB_C16_L3_VPN_Access_Register.sh.
Abbreviation in the view: TCP (Transmission Control Protocol). Keys are laboratory examples.

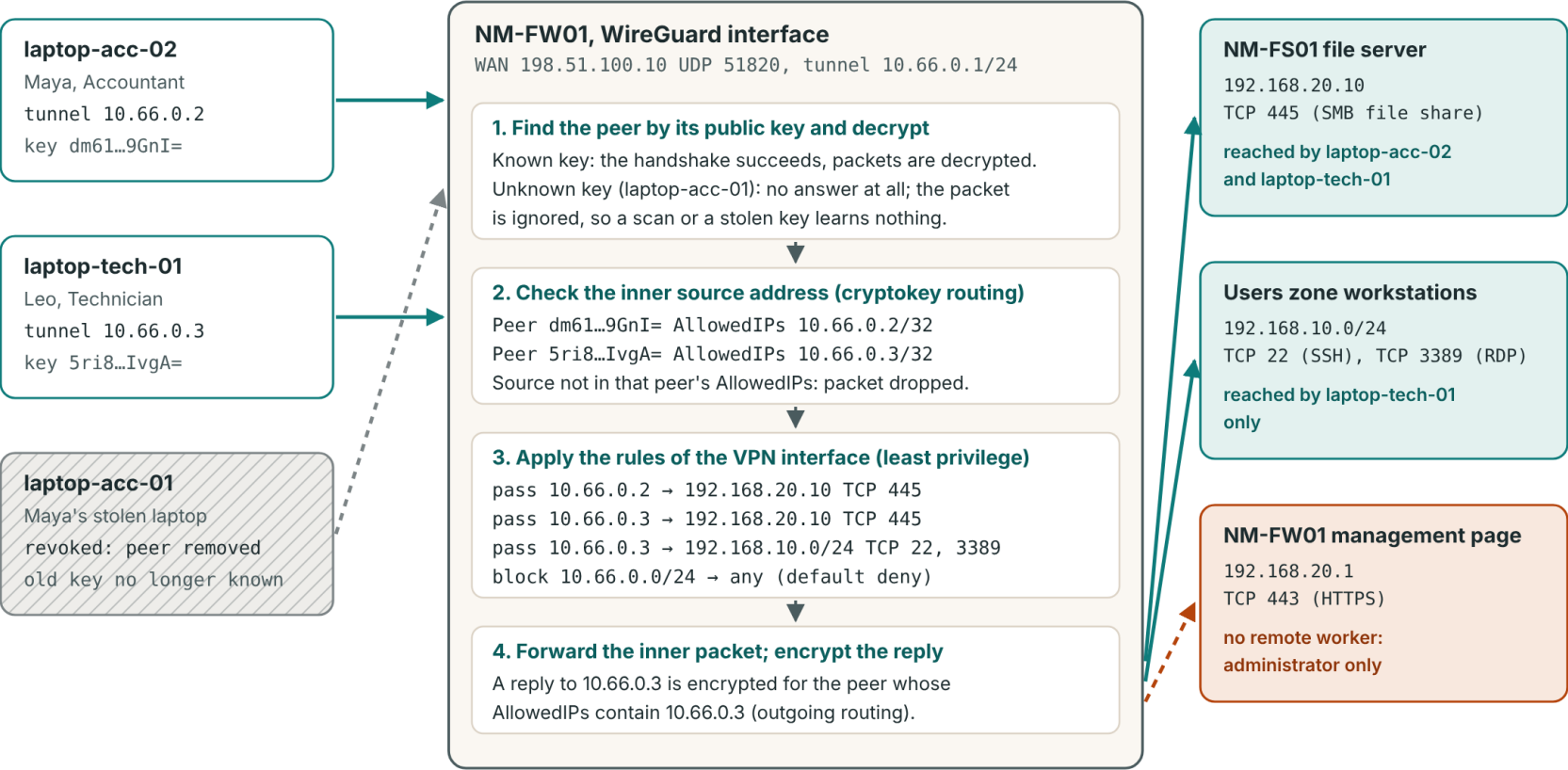
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Cryptokey routing and the tunnel rules

How NM-FW01 decides: cryptokey routing first, firewall rules second

Each remote device has its own key pair and its own tunnel address. Revoking a device means removing its peer.



Order of the decisions

- 1 Unknown key: no answer.
- 2 Inner source not in the peer's AllowedIPs: dropped.
- 3 Least-privilege rules of the tunnel interface.
- 4 Default deny for everything else.

Figure 3.1. Abbreviations in the figure: **UDP** (User Datagram Protocol), **TCP** (Transmission Control Protocol), **SMB** (Server Message Block), **SSH** (Secure Shell), **RDP** (Remote Desktop Protocol), **HTTPS** (Hypertext Transfer Protocol Secure).



LAB_06 preview: Class 17, Saturday, October 10

1 Part 1: remote-access tunnel

- Tunnel on NM-FW01; WireGuard recommended, alternatives stated in writing.
- Keys, users, allowed networks for NM-RW01 (10.66.0.2).
- Connection verified through 198.51.100.10.

2 Part 2: encryption and validation

- One folder or volume encrypted (data at rest).
- **HTTPS** (Hypertext Transfer Protocol Secure) with a self-signed certificate on one internal service.
- Evidence: tunnel traffic encrypted; service answers only over **HTTPS** (Hypertext Transfer Protocol Secure).

✓ Before Class 17

- NM-FW01 from LAB_05 runs; take a snapshot.
- NM-LX01 ready to clone into NM-RW01; **NAT** (Network Address Translation) Network NM-INTERNET exists.
- Manual: empty Chapter 6, headings 6.1 to 6.6.

Deliverable: Chapter 6 of the Final Project (2 points of 15) and of the Course Procedure Manual (1 point of 10), sections 6.1 Objective to 6.6 **AI** (Artificial Intelligence) disclosure.

Rule for tools that generate text (Legal Notice, section 11): state the **LLM** (Large Language Model) and the exact prompts. The model drafts; the analyst decides; the analyst's name goes on the record.



Official sources (1/4)

Product documentation

WireGuard: project page, design goals

<https://www.wireguard.com/>

WireGuard: protocol and cryptography

<https://www.wireguard.com/protocol/>

WireGuard: quick start, key commands

<https://www.wireguard.com/quickstart/>

WireGuard: official clients

<https://www.wireguard.com/install/>

WireGuard: whitepaper

<https://www.wireguard.com/papers/wireguard.pdf>

OpenVPN: community project

<https://openvpn.net/community/>

OpenVPN: version 2.6 manual

<https://build.openvpn.net/man/openvpn-2.6/openvpn.8.html>

Netgate: pfSense project site

<https://www.pfsense.org/>

Netgate: tunnel types of pfSense

<https://docs.netgate.com/pfsense/en/latest/vpn/index.html>

Netgate: choosing a tunnel type

<https://docs.netgate.com/pfsense/en/latest/vpn/selection.html>

Netgate: WireGuard package

<https://docs.netgate.com/pfsense/en/latest/vpn/wireguard/index.html>

Netgate: WireGuard tunnel and peer settings

<https://docs.netgate.com/pfsense/en/latest/vpn/wireguard/settings.html>

Netgate: OpenVPN on pfSense

<https://docs.netgate.com/pfsense/en/latest/vpn/openvpn/index.html>

Netgate: mobile IPsec (Internet Protocol Security) styles

<https://docs.netgate.com/pfsense/en/latest/vpn/ipsec/mobile-choices.html>

Netgate: pfSense release list

<https://docs.netgate.com/pfsense/en/latest/releases/index.html>

Deciso: OPNsense project site

<https://opnsense.org/>

Deciso: tunnel technologies in OPNsense

<https://docs.opnsense.org/manual/vpnet.html>

Check any link with the VirusTotal scanner before opening it: <https://www.virustotal.com/gui/home/url>. A link is not an endorsement (Legal Notice, section 10).



Official sources (2/4)

Product documentation (continued)

Deciso: WireGuard Road Warrior Setup

Deciso: two-factor authentication

Deciso: OPNsense 24.1 release notes

Noise Protocol Framework

Linux kernel: WireGuard padding code

Oracle: VirtualBox

<https://docs.opnsense.org/manual/how-tos/wireguard-client.html>

https://docs.opnsense.org/manual/two_factor.html

https://docs.opnsense.org/releases/CE_24.1.html

<https://noiseprotocol.org/>

<https://github.com/torvalds/linux/blob/master/drivers/net/wireguard/send.c>

<https://www.virtualbox.org/>

Standards

RFC (Request for Comments) 8439: ChaCha20 and Poly1305

<https://www.rfc-editor.org/rfc/rfc8439>

RFC (Request for Comments) 7748: elliptic curves, X25519

<https://www.rfc-editor.org/rfc/rfc7748>

RFC (Request for Comments) 7296: Internet Key Exchange version 2

<https://www.rfc-editor.org/rfc/rfc7296>

RFC (Request for Comments) 4301: security architecture of the Internet Protocol

<https://www.rfc-editor.org/rfc/rfc4301>

RFC (Request for Comments) 1918: private address blocks

<https://www.rfc-editor.org/rfc/rfc1918>

RFC (Request for Comments) 791: Internet Protocol

<https://www.rfc-editor.org/rfc/rfc791>

RFC (Request for Comments) 9293: Transmission Control Protocol

<https://www.rfc-editor.org/rfc/rfc9293>

IANA (Internet Assigned Numbers Authority): port registry

<https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml?search=openvpn>

Check any link with the VirusTotal scanner before opening it: <https://www.virustotal.com/gui/home/url>. A link is not an endorsement (Legal Notice, section 10).



Official sources (3/4)

Guidance and frameworks

NIST (National Institute of Standards and Technology): glossary definitions

https://csrc.nist.gov/glossary/term/virtual_private_network

NIST (National Institute of Standards and Technology): Special Publication 800-77 Revision 1

<https://csrc.nist.gov/pubs/sp/800/77/r1/final>

NIST (National Institute of Standards and Technology): Special Publication 800-46 Revision 2

<https://csrc.nist.gov/pubs/sp/800/46/r2/final>

Canadian Centre for Cyber Security: ITSAP.80.101

<https://www.cyber.gc.ca/en/guidance/virtual-private-networks-itsap80101>

Canadian Centre for Cyber Security: home page

<https://www.cyber.gc.ca/en>

NSA (National Security Agency): press release, 2021

<https://www.nsa.gov/Press-Room/Press-Releases-Statements/Press-Release-View/Article/2791320/nsa-cisa-release-guidance-on-selecting-and-hardening-remote-access-vpns/>

Joint guidance, 2021 (document)

https://media.defense.gov/2021/Sep/28/2002863184/-1/-1/0/CSI_SELECTING-HARDENING-REMOTE-ACCESS-VPNS-20210928.PDF

CISA (Cybersecurity and Infrastructure Security Agency): alert, 2021

<https://www.cisa.gov/news-events/alerts/2021/09/28/cisa-and-nsa-release-guidance-selecting-and-hardening-vpns>

CISA (Cybersecurity and Infrastructure Security Agency): exploited vulnerabilities catalog

<https://www.cisa.gov/known-exploited-vulnerabilities-catalog>

CIS (Center for Internet Security): Critical Security Controls 8.1

<https://www.cisecurity.org/controls/v8-1>

CIS (Center for Internet Security): Control 12

<https://www.cisecurity.org/controls/network-infrastructure-management>

Check any link with the VirusTotal scanner before opening it: <https://www.virustotal.com/gui/home/url>. A link is not an endorsement (Legal Notice, section 10).



Official sources (4/4)

Guidance and frameworks (continued)

CIS (Center for Internet Security): Safeguard 12.7

<https://controls-assessment-specification.readthedocs.io/en/latest/control-12/control-12.7.html>

CIS (Center for Internet Security): Safeguard 6.4

<https://controls-assessment-specification.readthedocs.io/en/latest/control-6/control-6.4.html>

Law and privacy

Criminal Code, section 342.1

<https://laws-lois.justice.gc.ca/eng/acts/c-46/section-342.1.html>

Criminal Code, section 342.2

<https://laws-lois.justice.gc.ca/eng/acts/c-46/section-342.2.html>

Criminal Code, section 430

<https://laws-lois.justice.gc.ca/eng/acts/c-46/section-430.html>

Québec privacy act (Law 25)

<https://www.legisquebec.gouv.qc.ca/en/document/cs/P-39.1>

CAI (Commission d'accès à l'information du Québec)

<https://www.cai.gouv.qc.ca/>

PIPEDA (Personal Information Protection and Electronic Documents Act)

<https://laws-lois.justice.gc.ca/eng/acts/P-8.6/>

Privacy Commissioner of Canada

<https://www.priv.gc.ca/en/>

GDPR (General Data Protection Regulation): Regulation (EU) 2016/679

<https://eur-lex.europa.eu/eli/reg/2016/679/oj>

Directive 2013/40/EU

<https://eur-lex.europa.eu/eli/dir/2013/40/oj>

CFAA (Computer Fraud and Abuse Act): 18 U.S.C. §1030

<https://uscode.house.gov/view.xhtml?req=granuleid:USC-prelim-title18-section1030&num=0&edition=prelim>

Spain: Ley Orgánica 10/1995

<https://www.boe.es/buscar/act.php?id=BOE-A-1995-25444>

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Questions?

Next class: **LAB_06, both parts**, Saturday, October 10, 2026.

After the break: videos, code runs, flashcards and practice questions on your page.

Bring NM-FW01 from LAB_05, working, with a snapshot.

